

Software Studio

軟體設計與實驗

Midterm Project Chatroom

Hung-Kuo Chu

Department of Computer Science
National Tsing Hua University

CS2410



Goal

- Create your chatroom app with Firebase.
- Your web page must meet all requirement in **basic** components then you will get **50%**
- Add some **advance** components and **bonus** components in your web page will get at most **45%**
- Describe **the functions of your website** and **how to use** in **README.md**
- You ought to explain
 1. how to operate your web,
 2. how to setup your project locally **STEP BY STEP**
 3. if TAs can't find the function, you won't get the points.
 4. If TAs can't setup locally, **-5** points



Basic components

- Membership Mechanism (5%)
 - Email Sign Up
 - Email Sign In
- Host your Firebase page (5%)
 - Use **Firestore** to host your web page and verify that it is functioning properly
- Database read/write (5%)
 - To read/write your membership data or other useful data, do it in **authenticated** way



Basic components

- RWD (5%)
 - Check your website working fine on different size device
 - All components must remain visible — nothing should disappear
 - If the layout requires scrolling on smaller screens, points will be deducted
- Git (5%)
 - Use Git to do version control
 - Commit regularly (not just on the last day!)



Basic components

- Chatroom **(25%)**
 - Can create **private chatrooms** to chat with other members register in this website
 - Other members can see your messages in chatroom
 - Load all history message of current chatroom
 - Make sure your chatroom can invite new members, like group chat in **Messenger**
 - Take **Messenger's** chatroom logic as reference if you are confused in implementation



Advanced components

- Using framework (React or others) (5%)
- Sign Up/In with Google or other third-party accounts (1%)
- Add Chrome notification (5%)
 - TA use “Chrome” for verification
 - Should notify with “unread” message instead all messages
- Use CSS animation (2%)
 - <https://www.youtube.com/watch?v=bjUoQbSJDJs>
 - Button hover is not an animation!
- Deal with problems when sending code (2%)
 - `<script>alert(“example”);</script>`
 - `<h1>example</h1>`



User Profile(10%)

- The user profile should be implemented as either a **modal** or a **separate page**
- It must include the following **editable** and **savable** fields:
 - Profile picture
 - Username
 - Email
 - Phone number
 - Address
- Need to show either **email, username or profile picture** in the chatroom



Message operation(10%)

- **Unsend message** (Only can unsend the message which send by yourself)
- **Edit message** (Only can edit the message which send by yourself)
- **Search message** (Can search for all message)
- **Send image** (Also can unsend the image which sent by ourself)



Bonus Components

- Bonus Components **(at most 10%)**
 - Chatbot **(2%)** (Must use ChatGPT or gemini api)
 - Block User **(2%)**
 - Send gif from Tenor API **(3%)**
 - Send the emoji to the message **(3%)**
 - Reply for specify message **(6%)**
 - Send custom sticker into chatroom **(10%)**



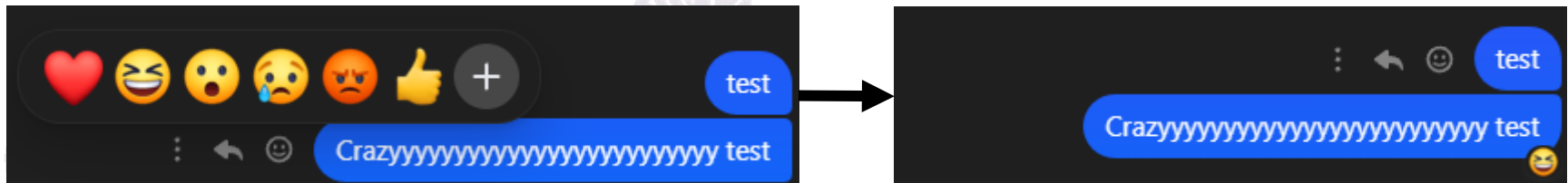
Block user(2%)

- When User A blocks User B, User B can't send direct messages to User A anymore.
 - If they have chat history, the chat UI must display a warning notification indicating that they can no longer chat.
- When User A (blocker) and User B (blocked) are in the same group chat, their messages should be mutually hidden.



Message emoji(3%)

- Can send the emojis to the message
- The number of available emojis can be decided by yourself
- User also can unsend their sent emoji
- If many people send the emoji to the same message, you can decide how many to show by yourself.(All of them or only few?)



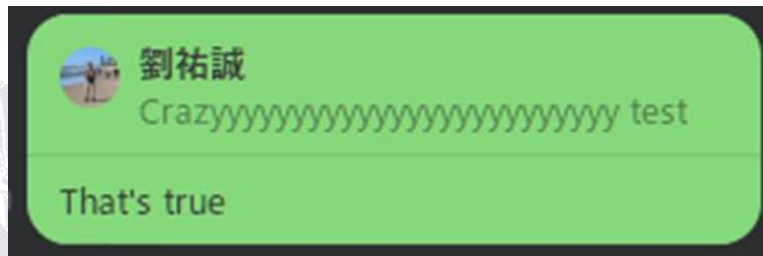
Reply for specify message(6%)

- User can reply for specify message (include the message send by himself/herself) and be sure have the clear UI to show the replied message and original message. (3%)
- Need to show the replied message above the text input when user is typing the response (1%)
- Clicking a replied message should trigger a visual highlight on the original message to indicate focus.(1%)
- Clicking a replied message should scroll the chat to the original message if it's out of view. (1%)

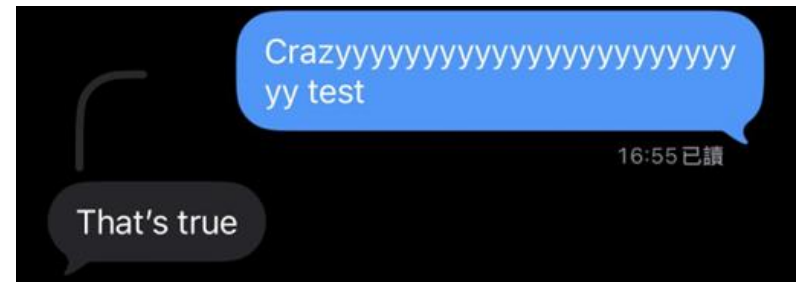


Reply for specify message

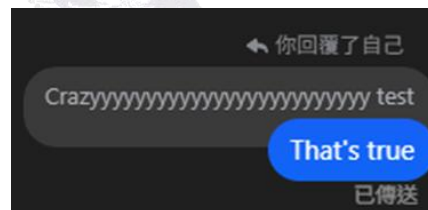
- Be sure have the clear UI to show the replied message and original message, following are few example. You also can design the style by yourself.



Line



iMessage

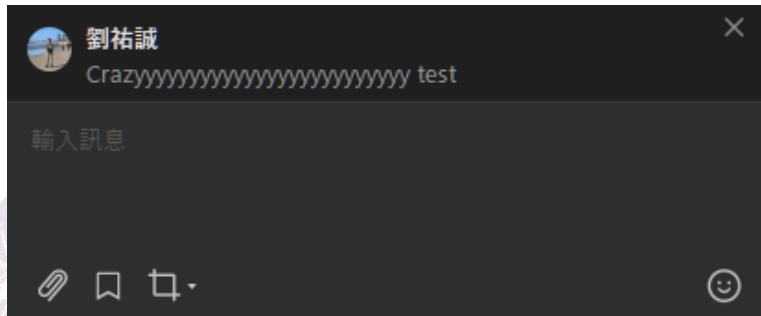


Messenger



Reply for specify message

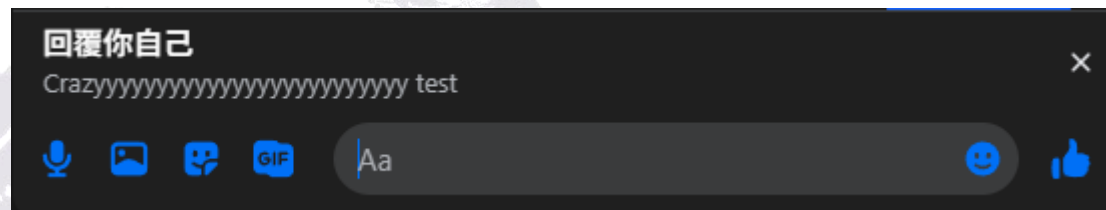
- Need to show the original message above the text input when user is typing the response



Line



iMessage



Messenger



Reply for specify message

- Clicking a replied message should trigger a visual highlight on the original message to indicate focus.



Line

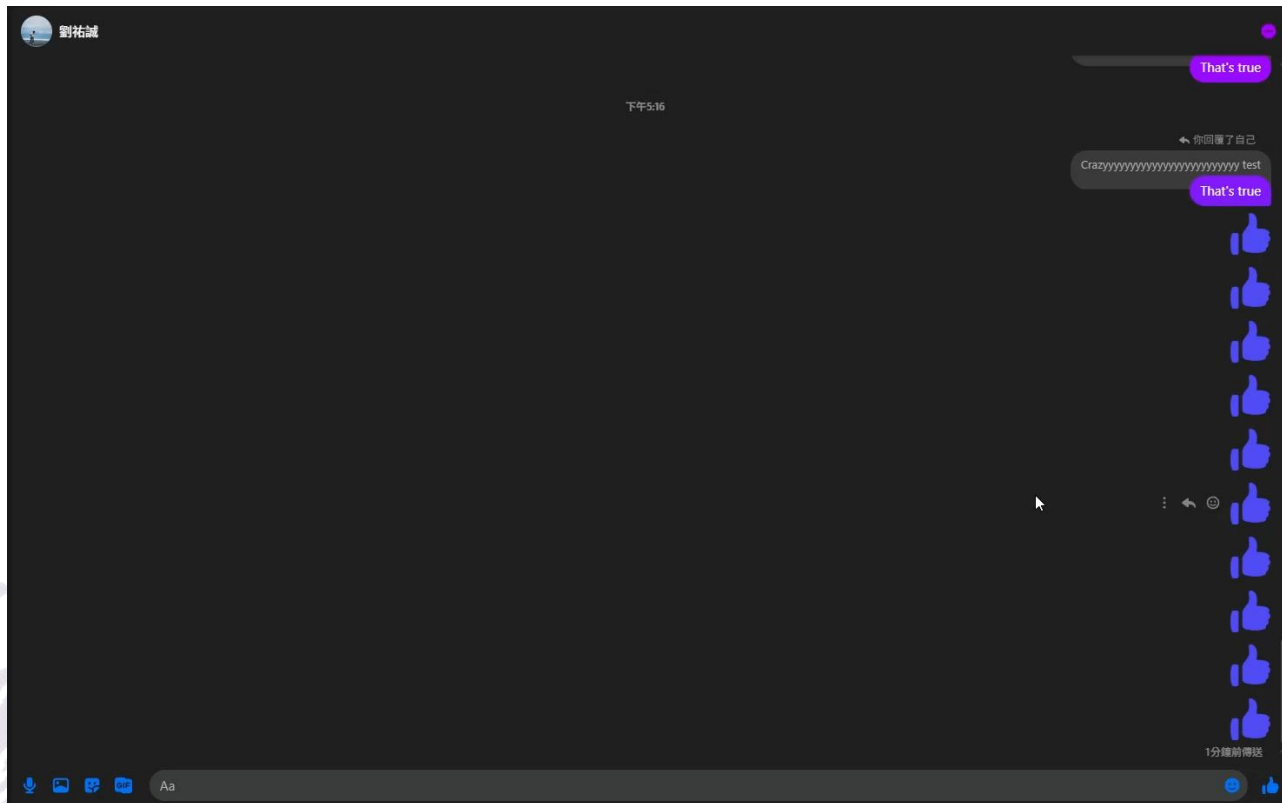


Messenger



Reply for specify message

- Clicking a replied message should scroll the chat to the original message if it's out of view.

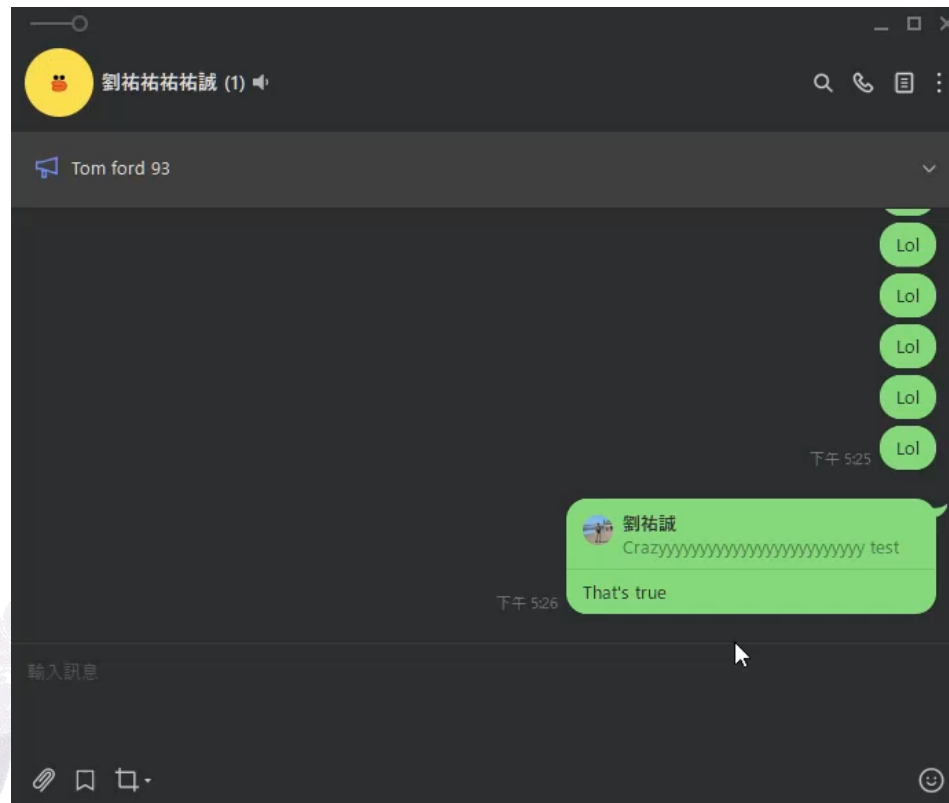


Messenger



Reply for specify message

- Clicking a replied message should scroll the chat to the original message if it's out of view.

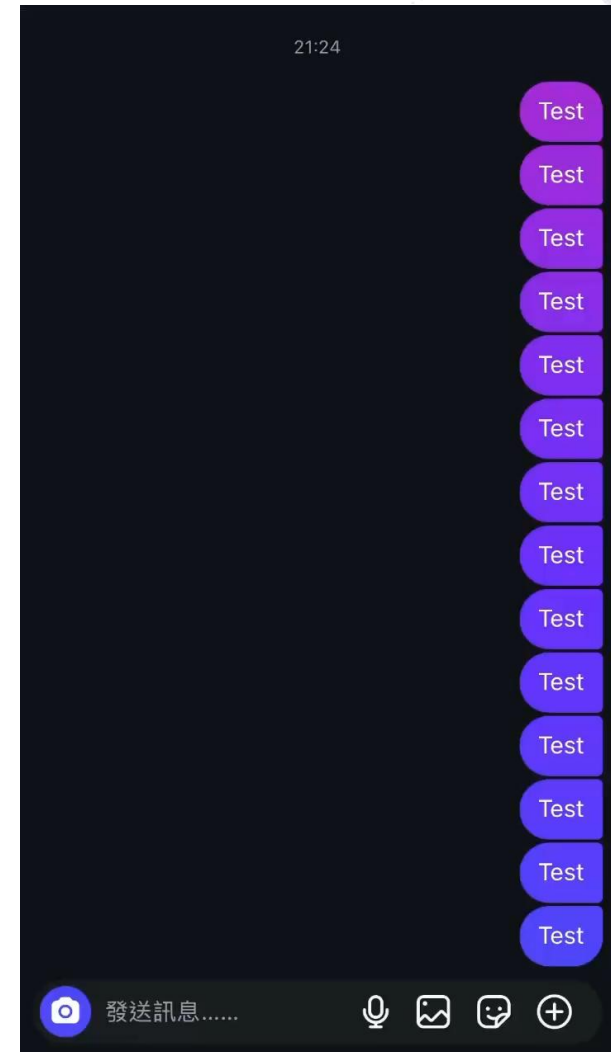


Line



Custom stick message(10%)

- IG provide a funny functionality in recent update; user can draw the custom sticker directly in the chatroom.
- We ask you to implement it in your chatroom



Custom stick message(10%)

Request:

- All UI of canvas should appear after the user click the drawing button.
- Can't edit the stick after user sends it
- The chatroom box will become canvas when user is drawing the custom stick
- You can send the stick as normal image if u can't keep it at the correct position, but the score of third criteria won't be given

Score criteria

- Multiple colors to choose (2%)
- Multiple brush to choose (2%)
- Should show the stick at the position where user draw it (4%)
- The custom stick can be unsent (2%)



Scoring

Item	Score
Basic components	50%
Advance components	35%
Total Completeness (subjective)	10%
Bonus components	10%
Total	105%

NOTE: README (-5), SOP (-10)



AI Usage

If you utilize AI tools during development, you must include a report titled **AI_reference.pdf** in the **root directory** of your project. The report must include:

1. **AI Tool(s) Used:** Specify the model (e.g., ChatGPT, Gemini...)
2. **Scope of Usage:** For every segment of code generated or assisted by AI, provide:
 - Location: File name and specific line numbers (e.g., App.js, lines 45-82).
 - Prompt & Response: The exact prompt you used and the AI's output (Screenshots are highly recommended).
 - Refinement & Explanation: Show your modified version of the code and provide a brief explanation of why you made those changes and how the logic works.
3. **Statement of Non-Usage:** If you did not use any AI tools, simply state: "No AI tools were used in this assignment." inside the PDF.

We strongly suggest that you thoroughly analyze any code produced by AI and rewrite it using your own logic, in case that it may be flagged as plagiarism regardless of the documentation.



Reminder

- Deploy your web page to **Firestore page**, and ensure it works correctly.
 - Your main page should be named as “index.html”
- Upload source code to FTP.
- Compress files into **Midterm_Project_學號.zip** then upload
 - index.html, .css, .js, README.md, etc.
 - **Do not add node_modules into zip file.**
(-5 if you upload node_modules)
 - If you upload the files again, please change the filename become **Midterm_Project_學號_v?.zip** (v? -> which version)
- **MD5 checksum** (if you didn't do this → **-10%**)
- **FIRM deadline: 2026/05/07 23:59** (commit time)
- Upload your **MD5** and **web link** and **GitHub Url** to eeclass



作業規則

- 程式碼嚴禁抄襲，抓到抄襲者，抄襲與被抄襲者

直接當掉！直接當掉！直接當掉！

直接當掉！直接當掉！直接當掉！

...

- 繳交期限為三個星期，不得遲交。
- 沒有屍體分數！
- 評分方式以日後公布為主。



如何不被判定抄襲？

- 參考相關網站的程式碼、透過生成式AI產生的程式碼，會因為來源相同、prompt相近導致被Moss判定為抄襲（例如function名稱相同）。
- 因此建議先理解網站\生成式AI\好友討論所產出的程式碼，再用自己的邏輯重寫，以降低被判定為抄襲的可能性。
- 若已知自己有涉及上述的手段來寫作業，建議先用MOSS比對自己的程式碼與參考的資料來源的相似度，確保比例低於20%，以降低被判定為抄襲的可能性。
- 關於Moss的申請與使用，可以參考下面的連結。
<https://hackmd.io/@GjP3NF8qT0SXa2TjRxSKeQ/HJvqIGerE>



作業繳交規則

- 請**務必**透過 MD5 獲得作業checksum 後填入 google 表單。
- 如遇各種原因無法在作業期限前完成上傳，我們將比對 checksum。若 checksum 一致則不算遲交。
- **繳交作業以MD5為主**，若有多個 checksum 則**取時間最晚**的為主。
- [MD5 online generator](#)
- [MD5 checksum 登記表單](#)
- [MD5 checksum 登記查看](#)
- [MD5 使用方法](#)



作業繳交流程

請務必遵守以下規則：

1. 將作業打包壓成zip檔，檔名格式以及打包內容以當次作業公告為主。
2. 幫zip檔產生MD5，並填寫google表單。
3. 將zip檔上傳ftp。
4. 將MD5和網址繳交至eeclass。

**若違反任何一項繳交作業SOP，
一律扣作業總分10分！！！！**

貼心提醒：請同學守護zip檔至作業成績公告或期末。

